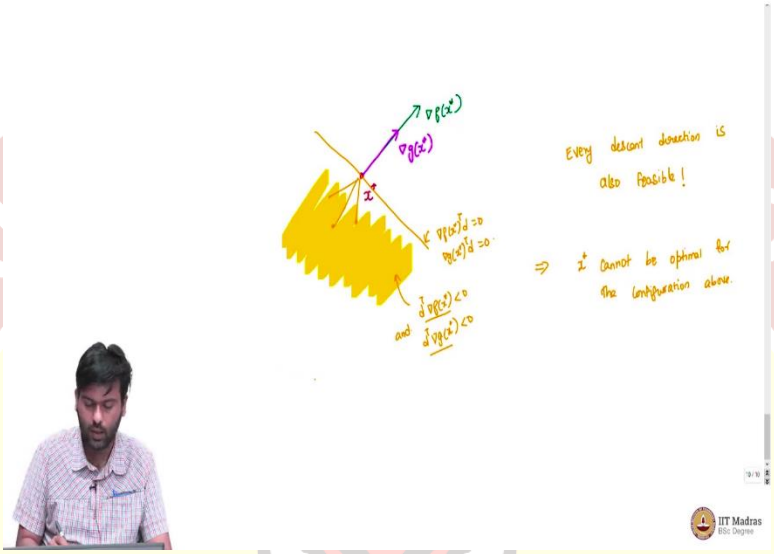


IIT Madras

ONLINE DEGREE

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Again, let us start with some point x star, let us see what are the possible choices, so now we want to understand how $\nabla f(x^*)$ and $\nabla g(x^*)$ should be related. Of course, you can pause and think here to see how these two should be related such that there is no descent direction, which is also feasible.

Let me go ahead and explain or take a different configuration for the gradients and see if that works. Let us, say if I took $\nabla f(x^*)$ and $\nabla g(x^*)$ to point in the same direction. Let us, see what happens then. This let us say this is and let $\nabla g(x^*)$ point in the same direction maybe not the same length, I mean the length does not really matter, it is only the direction that matters, maybe that this is delta g of x star, they are pointing the same direction.

Now, is this a configuration where there is no descent direction which is also feasible, again let us go through the motions. So, how would we find this? We need to find out where are the descent directions and the feasible directions in this case again we will see we will realize that this is this line where $\nabla f(x^*)$ transpose d equals 0 $\nabla g(x^*)$ transpose d equals 0 which means now if I look at this direction this has the property that any direction here which means that if I move here, here, here, no matter where I move, I am guaranteed that d transpose $\nabla f(x^*)$is less than 0 and d transpose $\nabla g(x^*)$is also less than 0.

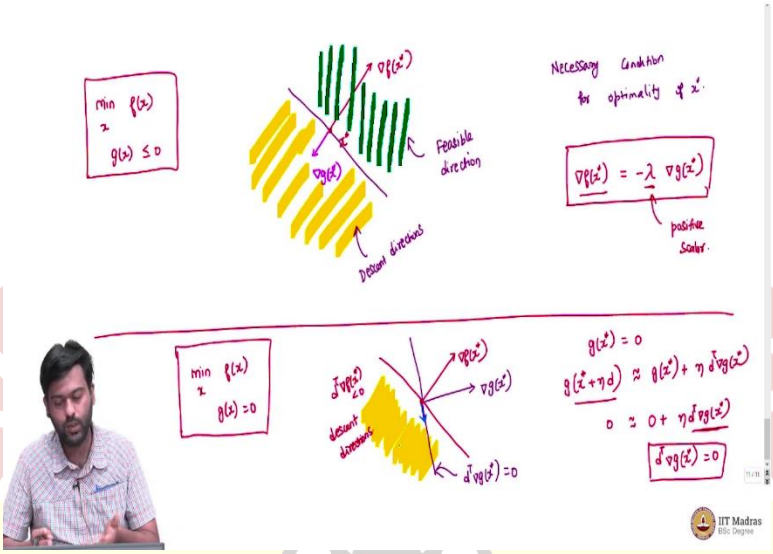
So, both of these hold, just clarify the I might sometimes use del f of x transpose d and sometimes d transpose $\nabla f(x^*)$ please do not get confused these are the same things, a transpose b same as b transpose a, and they are exactly the same numbers. So, now what is this mean?

This means that if I move in the yellow direction, so anywhere in the yellow space, if I end up in the yellow space, then I am both it is both the descent direction and a feasible direction which means that this is the in some sense the worst possible configuration because in this particular configuration every descent direction is also feasible, which is very bad.

So, because then in some sense, I have given you an x star which is really suboptimal x star, because I can move in any descent direction and I will still be feasible. So, at least in the previous case there were some directions where if you descended you may not be feasible, but all we need is a single direction to disprove my claim that x star is optimal here there are like a whole half space of directions, so which means that x star again cannot be optimal for this configuration for the configuration drawable.

Now, then, is there even a configuration where it will indeed be optimal? Well, by now you should have guessed what is the configuration, it is not too hard to see that maybe I will use a different page. So, it is not too hard to come up with the correct configuration now.

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Well, if you are at x star and your $\nabla f(x^*)$ points in this direction, but if you are $\nabla g(x^*)$ which could be of a different length potentially points in the opposite direction in other words $\nabla f(x^*)$ and $\nabla g(x^*)$ or anti-parallel this is delta g of x star, so then where are the descent directions? Well, the descent directions are here the yellow directions on this side of the purple line, these are the descent directions.

On the other hand, because your $\nabla f(x^*)$ is pointing to the opposite side, you were feasible directions are all going to be on the other side, this will be your feasible directions which means that the moment I have found a direction where I can descend will not be feasible, because the feasible directions are on the other side right. So, at least in this picture that is what that kind of tells us. Now, what can we say about this?

So, essentially finally, what we are saying is that a necessary condition for x star to be optimal is the following. So, the necessary conditions is not a sufficient condition and necessary condition for optimality of x star is that these $\nabla f(x^*)$ and del of g of x star should be anti-parallel. How can I make it precise enough in a simple formula?

Well, I can say that $\nabla f(x^*)$ equals some - lambda times $\nabla g(x^*)$, where this lambda is a positive scalar. So, there exists some positive scalar lambda such that $\nabla f(x^*)$ is not exactly - $\nabla g(x^*)$this lens could be different and that the length I mean it is the direction the same, but the length could be different and so you have a scaling factor here which is given by lambda.

Now, this is the condition that one would want to check this is a necessary condition for optimality of x star for a constrained optimization problem. So, remember our problem was minimize over x, f of x such that g of x is less than or equal to 0. So, this is where we have reached. Now, let me also quickly try and do what happens when you have minimize over x, f of x such that g of x equals to 0. So, this is like an equality constraint.

Now, what changes if you have a equality constraint one can ask that question. So, we can go over the same kind of argument to see when that will be a descent direction which is also feasible, but we just have to be careful of where the directions feasible directions are, this is again an intuitive picture, so you can again say that if let us say $\nabla f(x^*)$ was pointing in a particular direction let us say in this particular direction, nothing changes with respect to f because nothing has changed in the problem with respect to f the only g has changed.

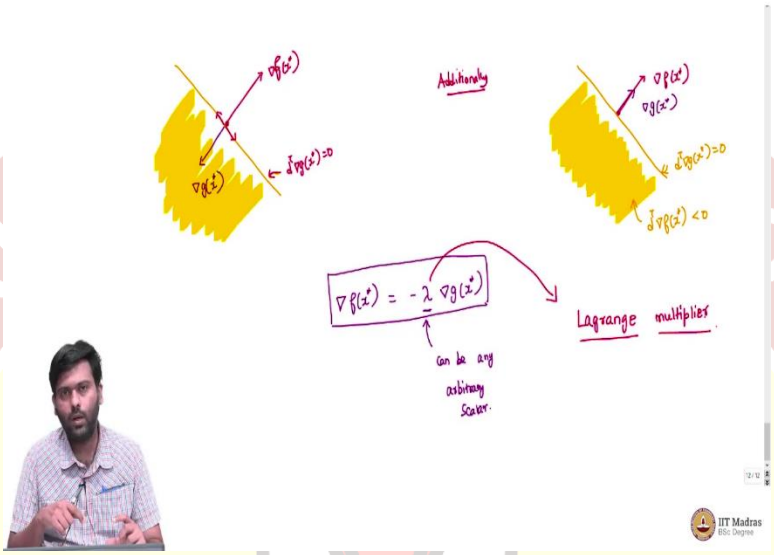
So, the descent directions will still be here, this is still the descent of I mean these are the directions where you can descent. So, 3 transpose $\nabla f(x^*)$ is less than 0 or descent directions. Now, the question is, where are the directions where you know you have feasibility? So, let us say you have you start with the g of x star which is kind of 0 which is feasible, and now you go to you make take a very small step which is you go to g of x star + some eta times d where eta is really small we know Taylor series would say that this is approximately g of x star + eta d transpose $\nabla g(x^*)$.

Now, if this also has to be feasible, well this is also 0, so this is 0 + eta times d transpose $\nabla g(x^*)$, so this is again an intuitive picture saying that if this condition has to hold then it is like saying that we want so move along the direction d such that which makes a dot product of 0 with the with the gradient with respect to g. So, which means that, let us say if the gradient with respect to g was pointing again in this direction, $\nabla g(x^*)$well for a small enough step you want to kind of move in a direction which is making a dot product of 0 with delta g of x star, where are those? They are here.

So, this exactly on this line, we will have $\nabla g(x^*)$equals 0. Now, on this line we can move anywhere and then the descent directions are all the yellow region which are a whole hub space. Now, is this configuration optimal? Of course, it is not optimal because I can always move in a small direction which is making a dot product of 0 with a gradient, but then it will take me to a descent direction,

so this is not an optimal configuration. So, now we are again asked the question, where is the optimal configuration?

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Well, now, the situation is slightly different from earlier because we do not have a whole half space of feaVsible regions, but then only like a line. Now, if you have again $\nabla f(x^*)$, now if let us say $\nabla g(x^*)$was opposite direction like how we had earlier for the case where you had an inequality constraint, let us say this was $\nabla g(x^*)$, well as before, so this is the feasible region I am sorry, this is the descent region and the feasible region is on the orange line.

So, the feasible sorry, should I use a different color. So, this is d transpose $\nabla g(x^*)$0, the orange line whereas the yellow line is the descent direction, which means that the moment I move along the yellow direction, I kind of go away from the orange line, so I cannot move in the orange line and still descent that is the point, if we move in this direction, it is also going to make a dot product of, also going to make a dot product of 0 with the del of f.

So, it is not really a descent direction, only when there is strictly less than 0 it becomes a descent direction which means that on the line I can still move and I will not be able to descend, so which means that this is a configuration which is in fact an okay configuration. So, you are which is exactly the same configuration that we had for an inequality constraint.

But then additionally, there is one other configuration which will also work the equality case which is which did not work in the inequality case, which is when not only the case when $\nabla f(x^*)$ and

del of g of x star anti-parallel, but even in the case when they are actually parallel. So, even if they are parallel let us say $\nabla g(x^*)$, now if I still ask the same question, well this is the case where this is d transpose $\nabla g(x^*)$is 0, it does not change, just because we are making a $\nabla g(x^*)$we are flipping it from the anti-parallel to parallel the line which makes a dot product of 0.

So, the line which is perpendicular remains exactly the same. And so, you can potentially have this configuration as well, where you will still will not be able to find a descent direction which is also feasible. So, that is the difference between the inequality constraint and the equality constraint. So, this is let me make it clear this is del of d transpose $\nabla f(x^*)$ is less than 0. So, which means that the moment we come out of this line we might descend, but then we might not be feasible. So, that is the whole point.

So, which means that there are two cases now, either g of del f g of x star is parallel or it is ant-parallel. So, now how do we formalize that? We can formulate that again in exactly the same way as we did before we can say that $\nabla f(x^*)$ equals - lambda $\nabla g(x^*)$this is what we had earlier and then we said that because we only allowed anti-parallel in the previous case we constrain lambda to be positive, here because we are allowing it to be either positive or negative, there is no constraint on the sign of lambda, we are simply going to say that lambda can be any arbitrary scalar.

So, it can be anything basically return nonzero arbitrary scalar. So, that as long as this condition is satisfied, then it is a necessary condition for the constrained optimization problem with an equality constraint. So, now this lambda has a special name and it is named after Lagrange famous scientist mathematician and it is called as Lagrange multiplier.

Now, what have we said so far? So, all we have said so far is we started with an unconstrained problem, we said that gradient descent would work there, but then the moment you introduce constraint gradient descent may not work. And then we asked when would an x star be optimal, what is the necessary condition, and the necessary condition that we have come up with is that $\nabla f(x^*)$ should be - lambda.

And in the case of an equality constraint lambda can be anything does not have any constraint, but then in the case of inequality lambda has to be exactly 0. Now, the question is fine, so this is an optimality constraint, but then how do I use this to actually solve a problem? So, how do I finally

get the answer by using this using this equation? So, that is a question that we want to answer, which is for which we will see an example a specific example with an equality constraint where you can use the method of Lagrange multipliers to solve the problem and that is what we will see next.